



Module 1 Day 1: Activity 11 CIA Triad

Activity Introduction

- Analyze the brief security scenarios below and identify which element of the CIA triad (confidentiality, availability, or integrity) each situation is concerned with.

Scenarios:

1. A hospital only allows authorized healthcare personnel within one department to access patient Personal Identifiable Information. When employees move to another department, they lose access.
2. A technology firm maintains an alternate site that is running at all times. Operations can be moved to this location in the event of a major disaster.
3. Employees need key cards in order to enter their company offices.
4. A company hashes their data files to monitor whether information has been tampered with.
5. Only authorized personnel at a company have write access to certain files. All other employees only have read access to these files.
6. A company employs redundant servers, which means that systems are duplicated. In the event of a malfunction, one server will step in for the other.
7. A company's network infrastructure uses load balancers, which distribute the "load" of tasks, such as file requests and data routing, to a variety of servers. This ensures that no single device is overburdened.
8. A cybercriminal uses an on-path attack to intercept wireless traffic from users.

9. A cybercriminal cracks a hashed message and changes its contents.
10. A cybercriminal launches a DDoS attack that floods a website with unwanted traffic from multiple computers and takes the site offline.